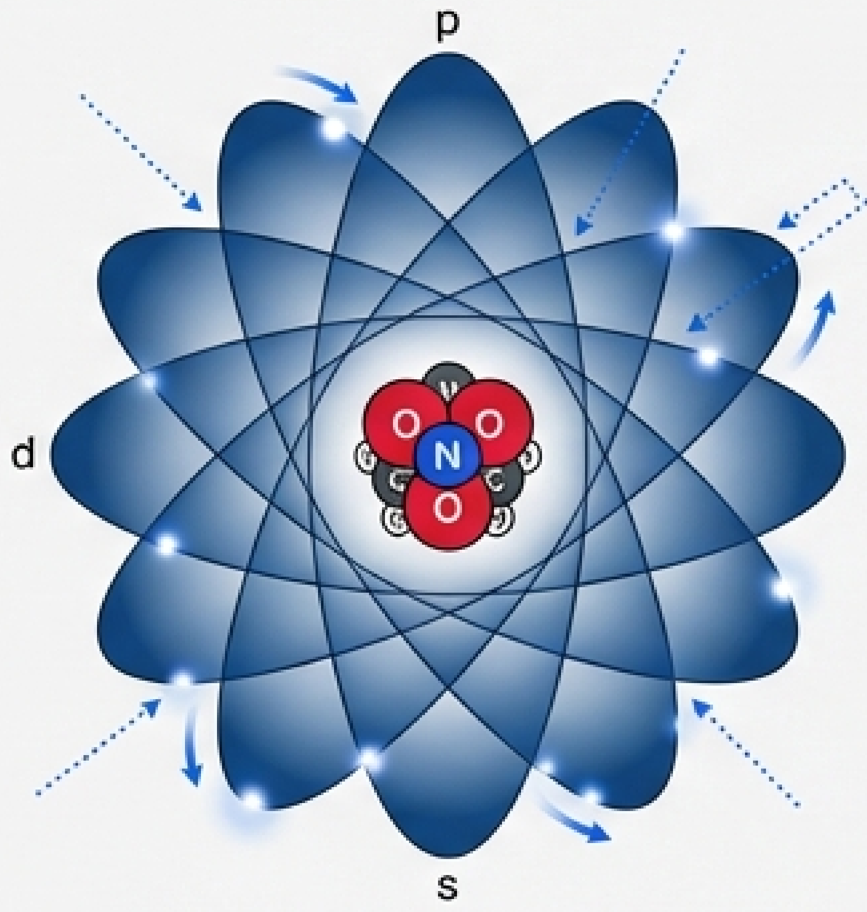
The Building Blocks of Biochemistry

A Visual Guide to Fractionating Life, from Atoms to Emergence.

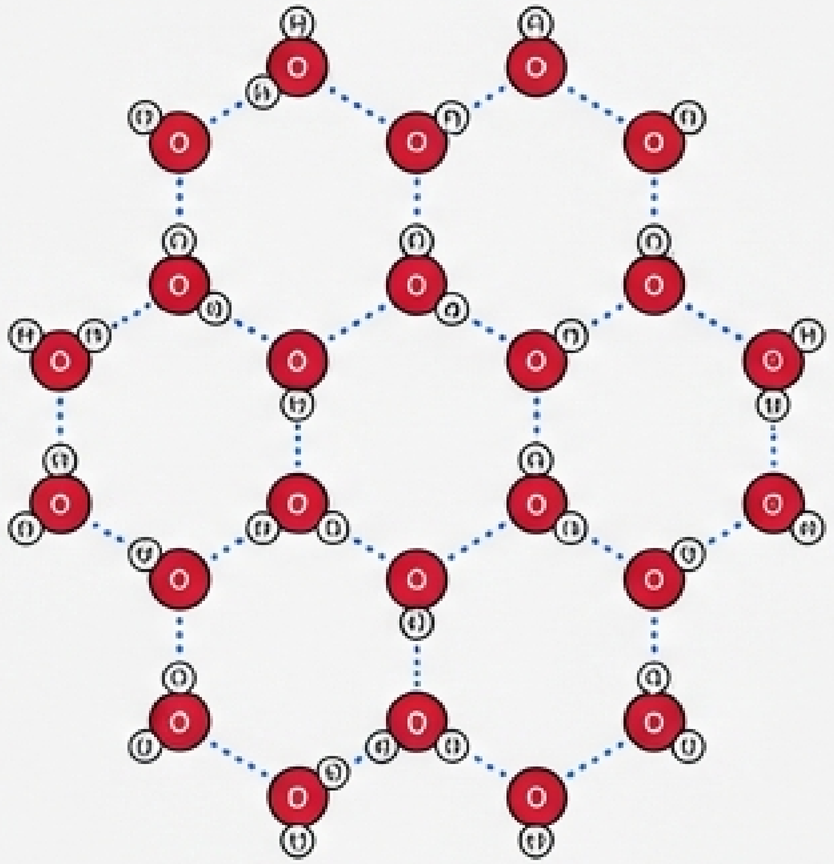
ATOM



ELECTRON CLOUD & NUCLEUS

- Protons & Neutrons (Nucleus)
- Electrons (Orbitals)
- Quantum Mechanics Foundation

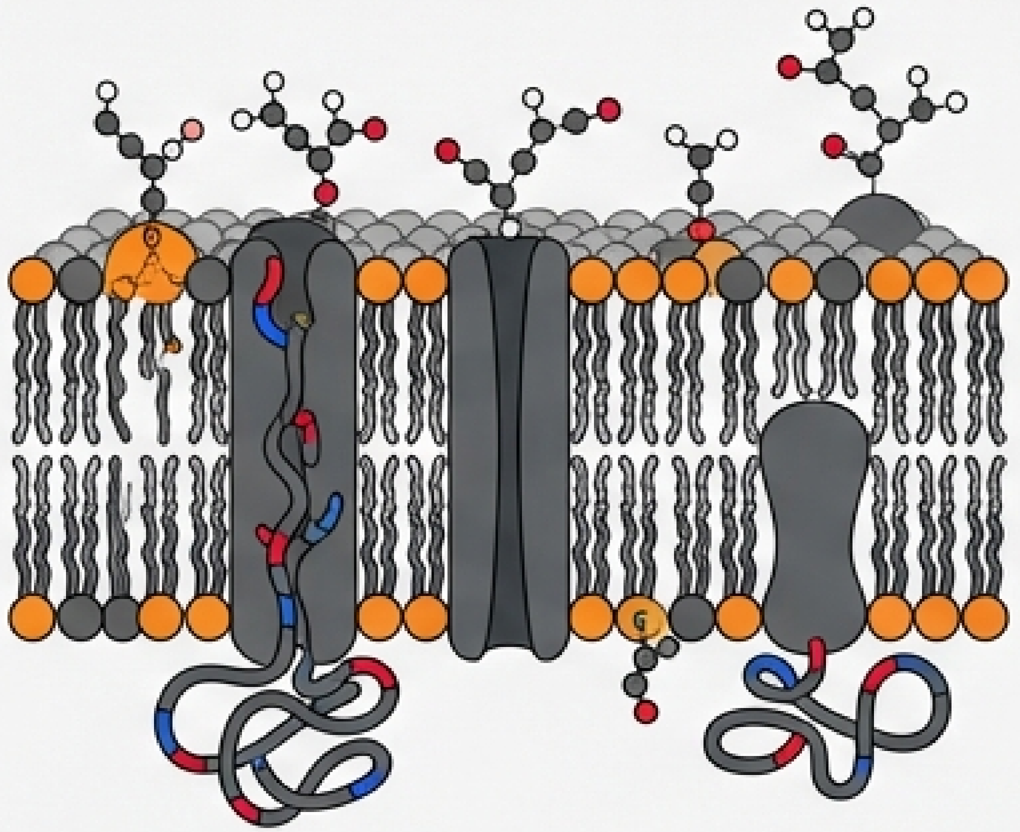
MOLECULE



WATER (H_2O) LATTICE

- Polar Covalent Bonds
- Hydrogen Bonding Network
- Cohesion & Adhesion Properties
- Universal Solvent

CELLULAR STRUCTURE



LIPID BILAYER & PROTEINS

- Phospholipids (Amphipathic)
- Hydrophilic Heads & Hydrophobic Tails
- Membrane Proteins (Transport, Signaling)
- Fluid Mosaic Model

Based on MIT 7.00x: The Secret of Life

The Atomic Inventory: The Big Six

Life uses a tiny fraction of the periodic table.

H
Hydrogen
1 Bond
Abundance: 63%

O
Oxygen
2 Bonds
Abundance: 24%

N
Nitrogen
3 Bonds
Abundance: 1.4%

C
Carbon
4 Bonds
Abundance: 10%

P
Phosphorus
5 Bonds

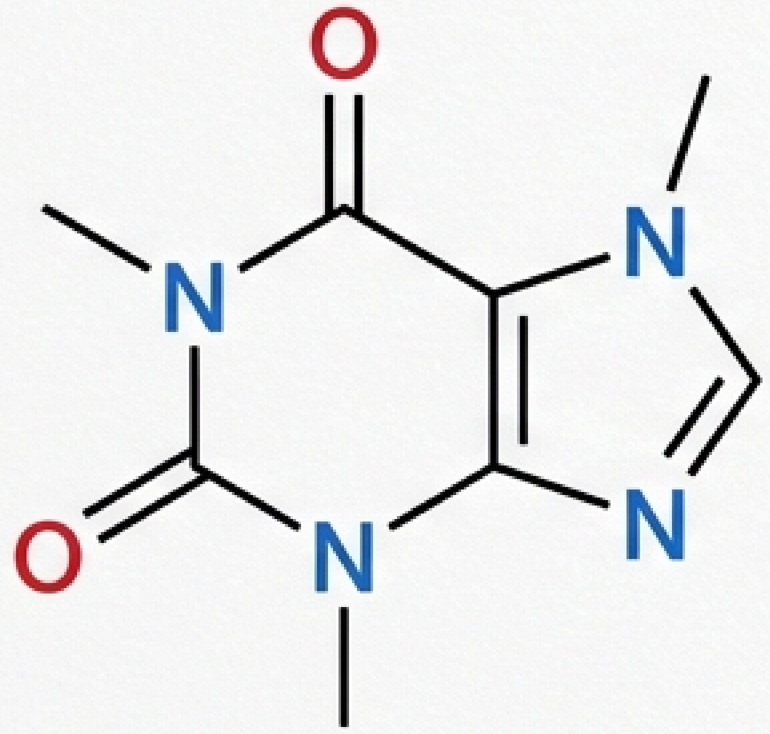
S
Sulfur
6 Bonds

Trace Ions

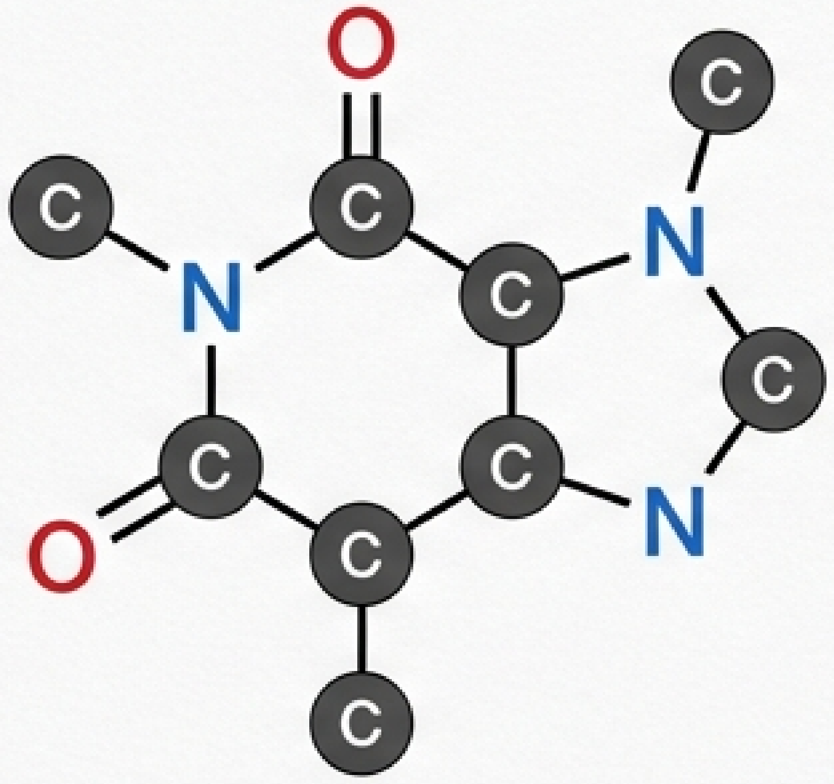
Na, K, Ca, Mg, Mn, Cl

Decoding the Shorthand

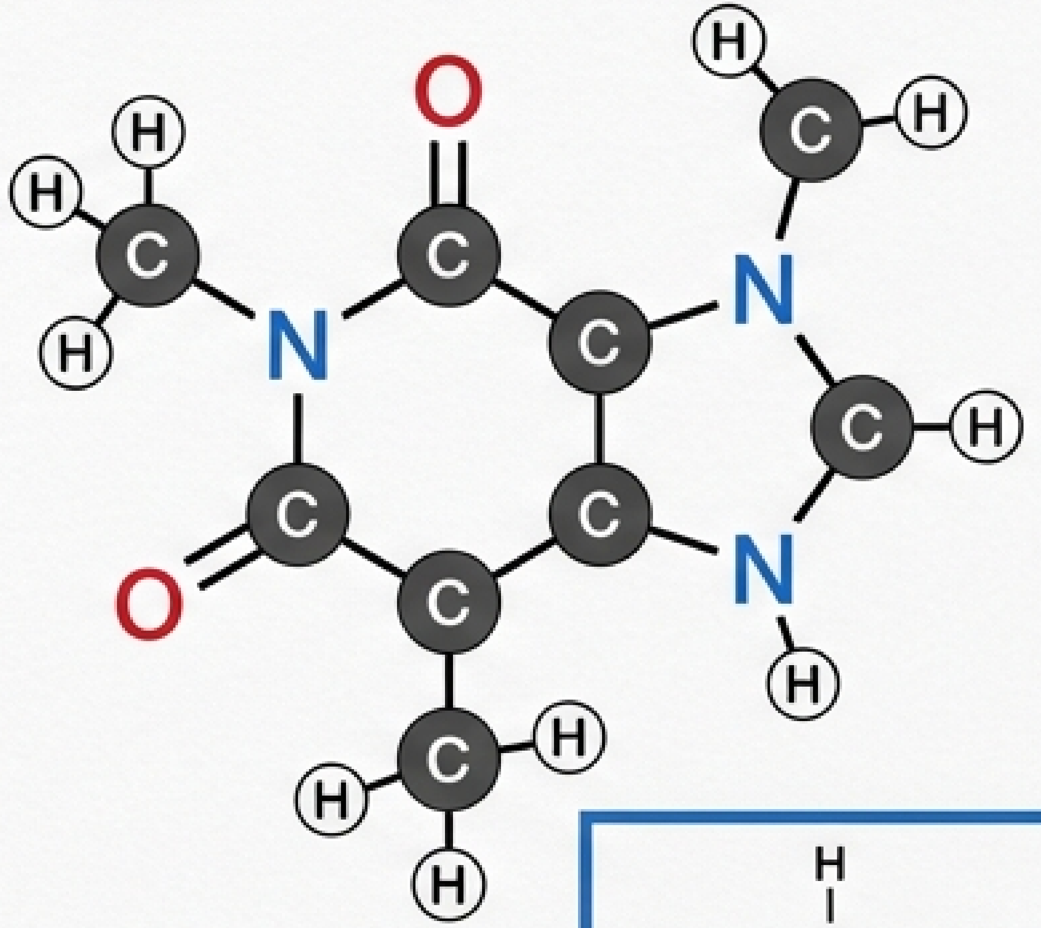
Skeletal View
Crimson Pro



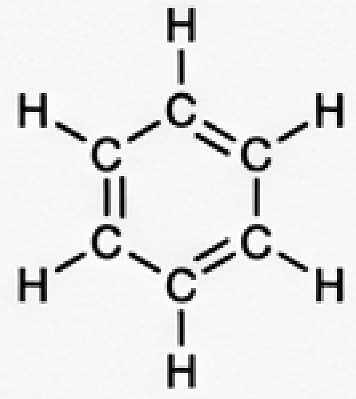
Rule 1: Carbons at Corners
Crimson Pro



Rule 2: Implied Hydrogens
Crimson Pro



Carbon always forms 4 bonds. If fewer are shown, fill the remainder with Hydrogens.
Crimson Pro



Check Yourself: Count the Hydrogens in this ring.
Crimson Pro

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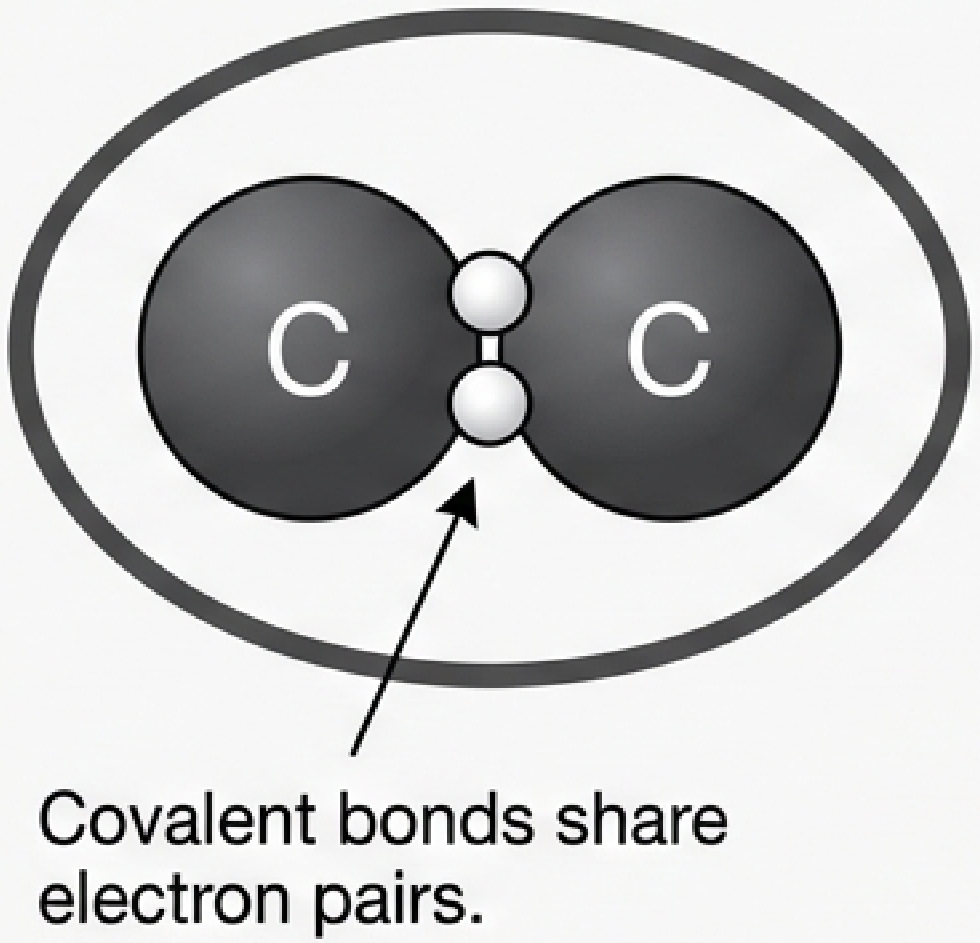
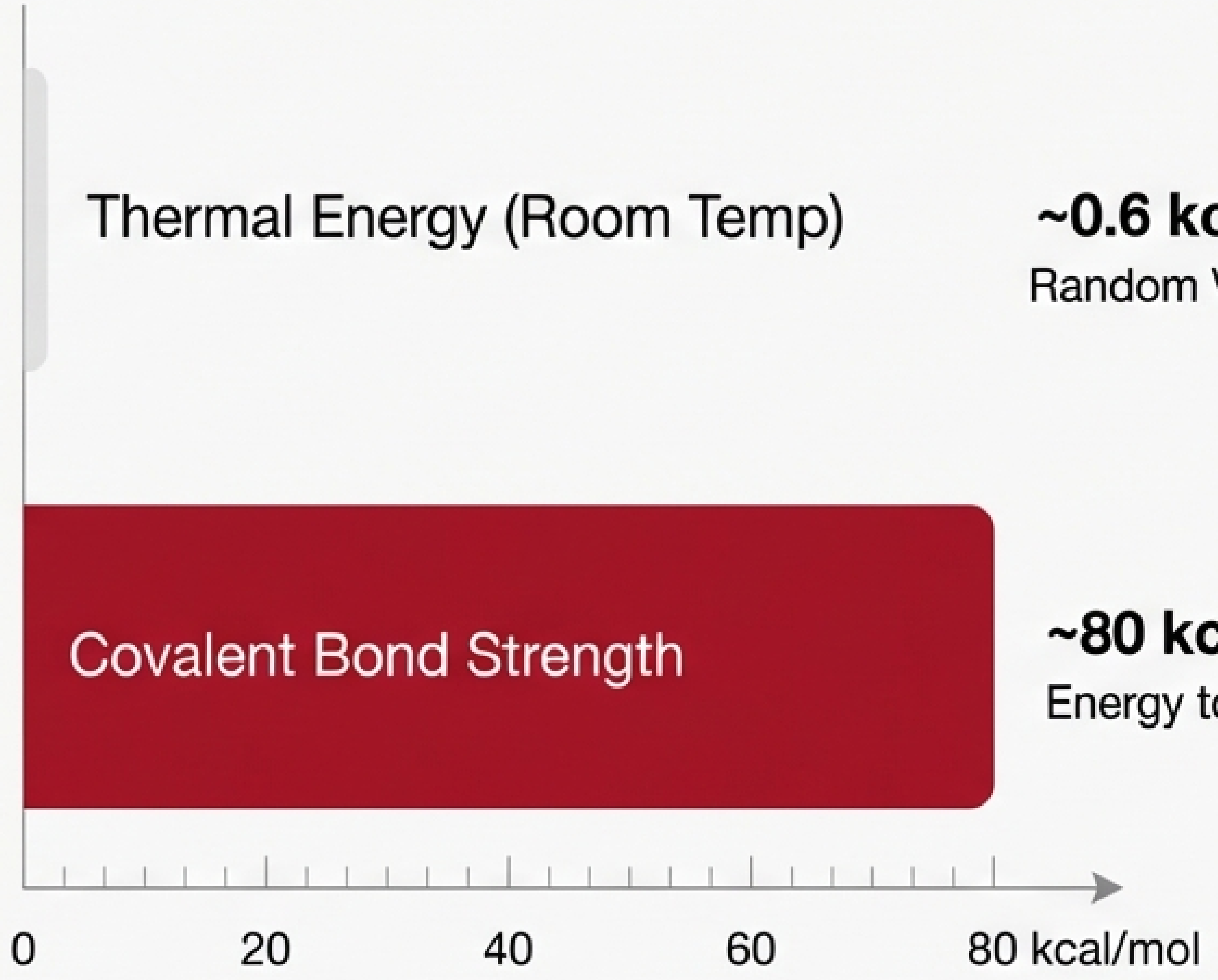
Covalent Bonds: The Permanent Glue

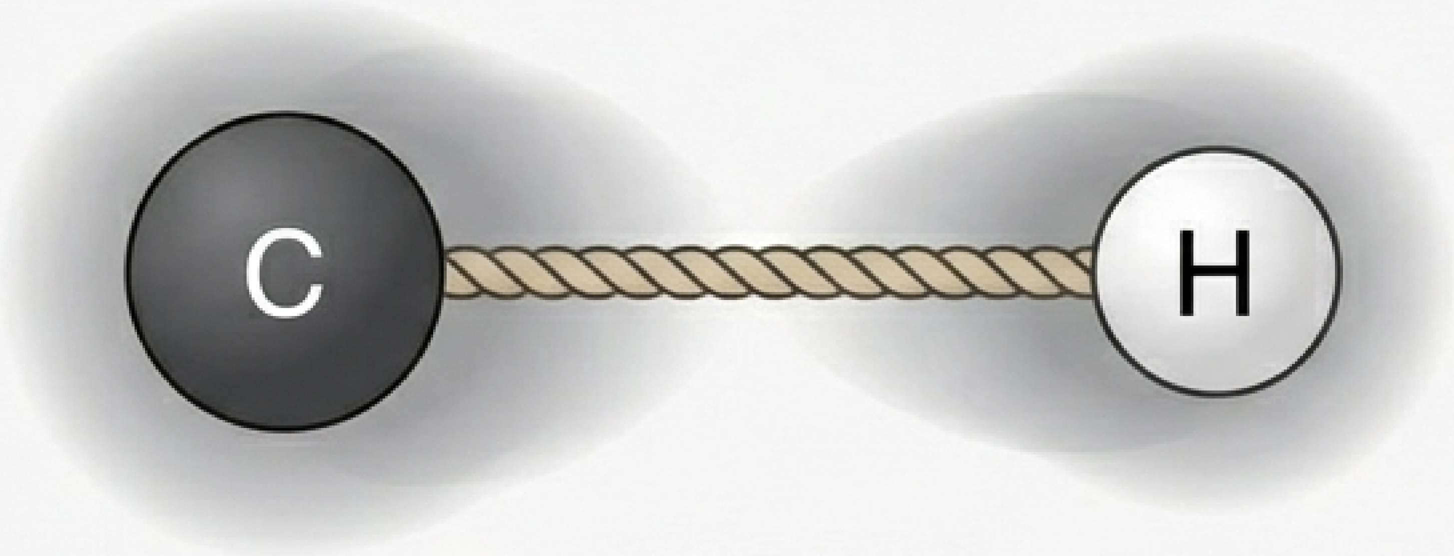




Stability: Random thermal fluctuations (0.6 kcal) are insufficient to break covalent bonds (80 kcal). Life structures do not fall apart spontaneously.

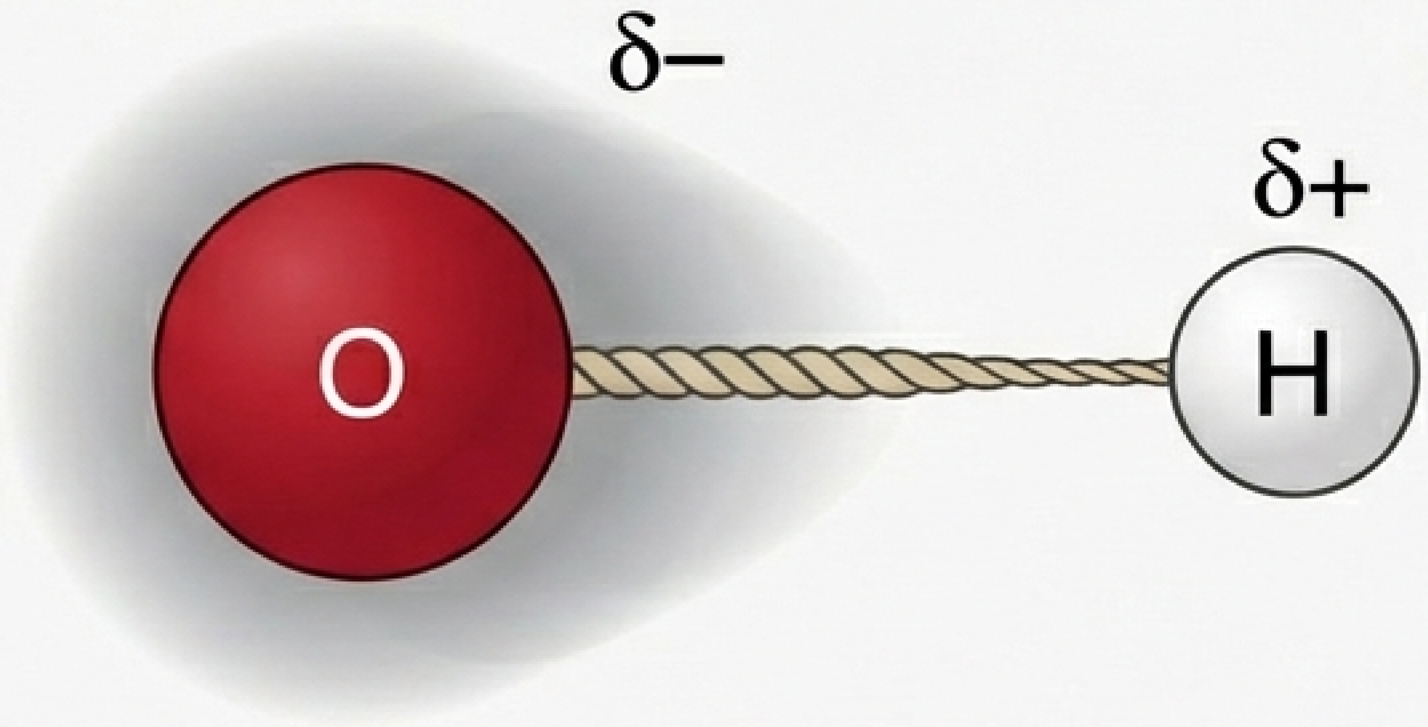
Electronegativity: The Atomic Tug-of-War

Non-Polar (C-H)
Crimson Pro



Equal Sharing
Electronegativity Difference < 0.5
Crimson Pro

Polar (O-H)
Crimson Pro



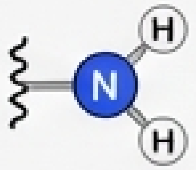
Unequal Sharing
Crimson Pro

Rule of Thumb: Greedy Atoms: Oxygen (O) & Nitrogen (N).
High Polarity = Bonded to O or N. Low Polarity = Bondded to C or H.
Crimson Pro

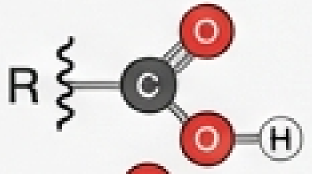
The Toolkit: Functional Groups Roups Reference

Hydrophilic (Polar / Water Lovers)

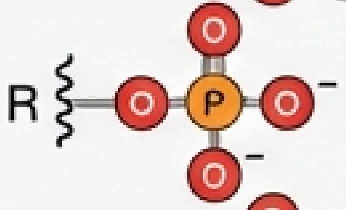
Hydroxyl (-OH)



Amino ($-NH_2$)



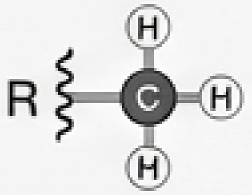
Carboxyl (-COOH)



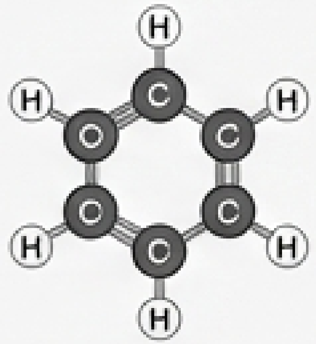
Phosphate ($-PO_4$)

Aldehyde (-CHO)

Hydrophobic (Non-Polar / Water Fearers)

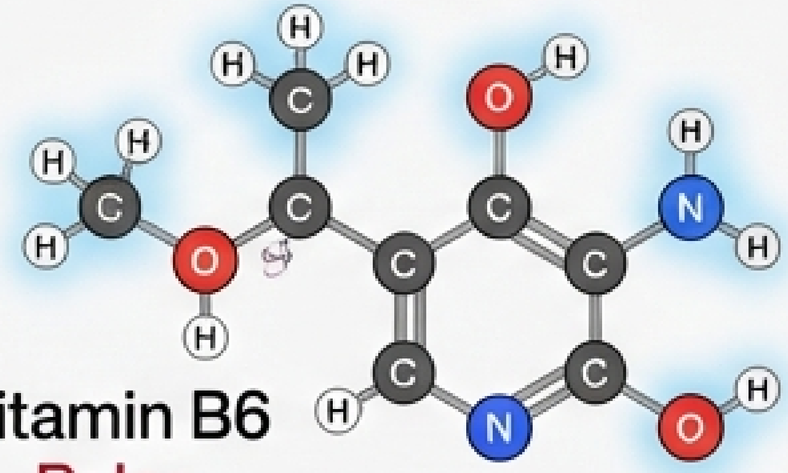


Methyl ($-CH_3$)

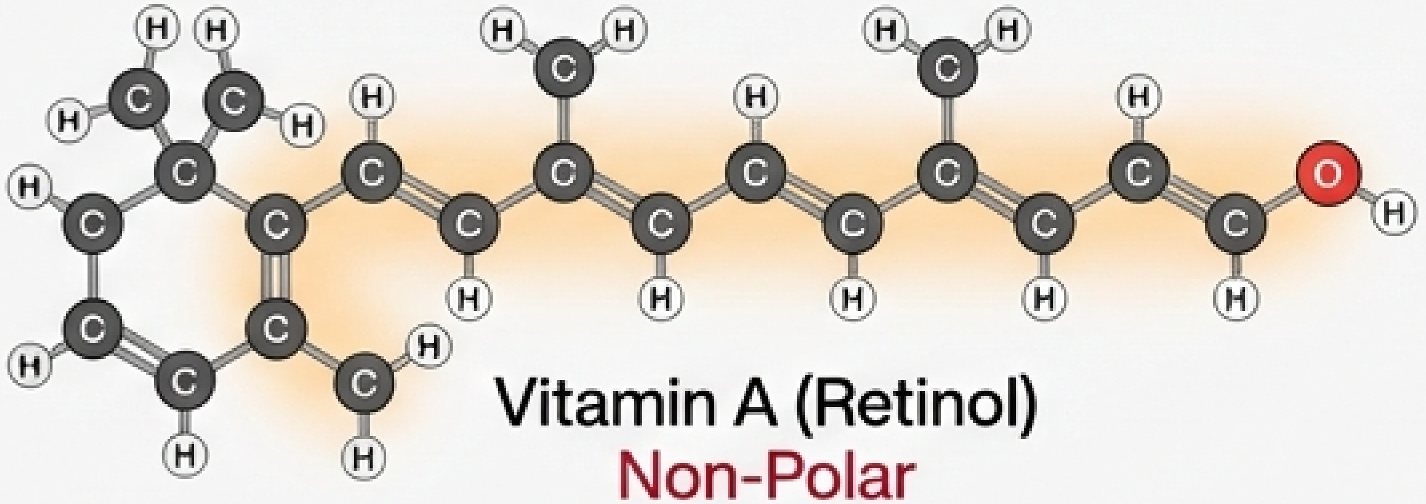


Phenyl Ring

Sulfhydryl (-SH)



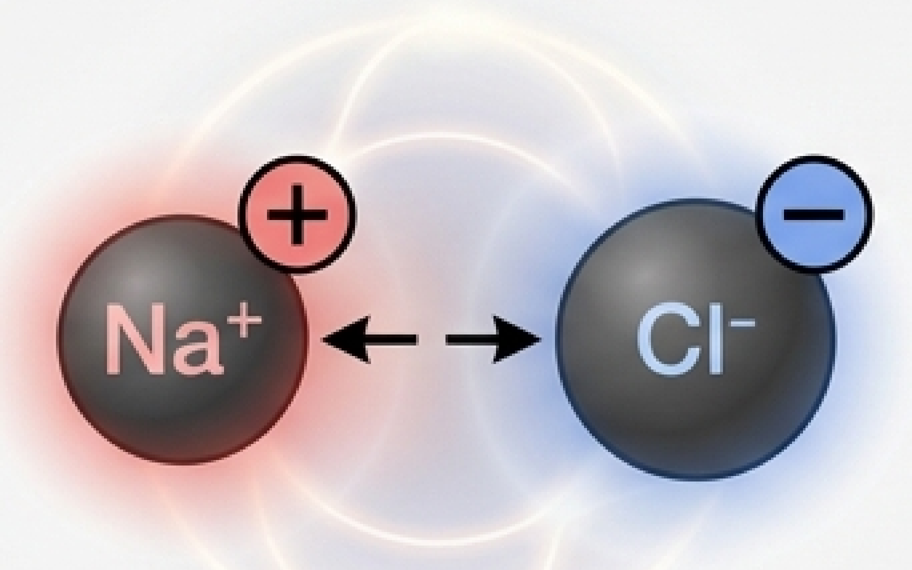
Vitamin B6
Polar



Vitamin A (Retinol)
Non-Polar

The Velcro of Biology: Non-Covalent Interactions

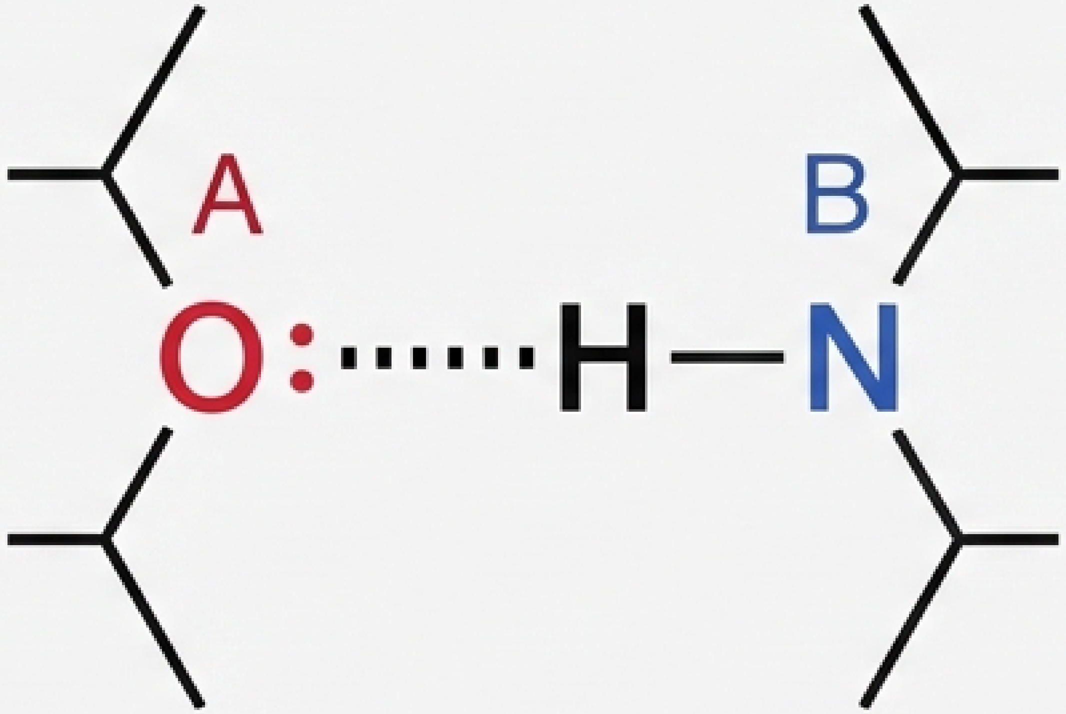
Ionic Bonds



Full Charge Attraction (+/-)

(Crimson Pro)

Hydrogen Bonds

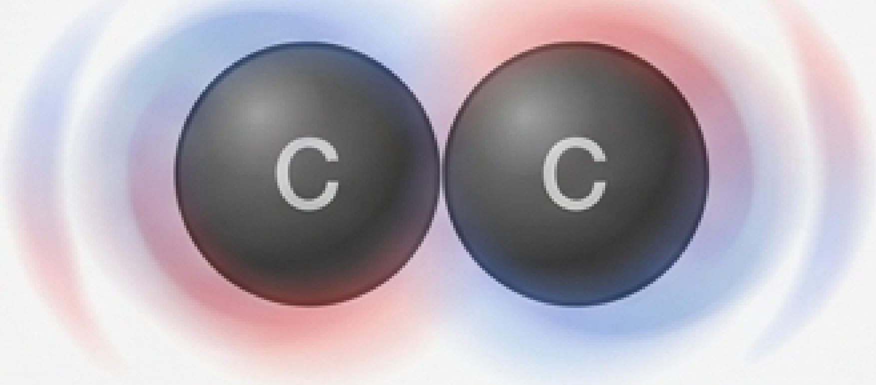


'Sandwich'

The Polar Sandwich (~5 kcal/mol)

(Crimson Pro)

Van der Waals

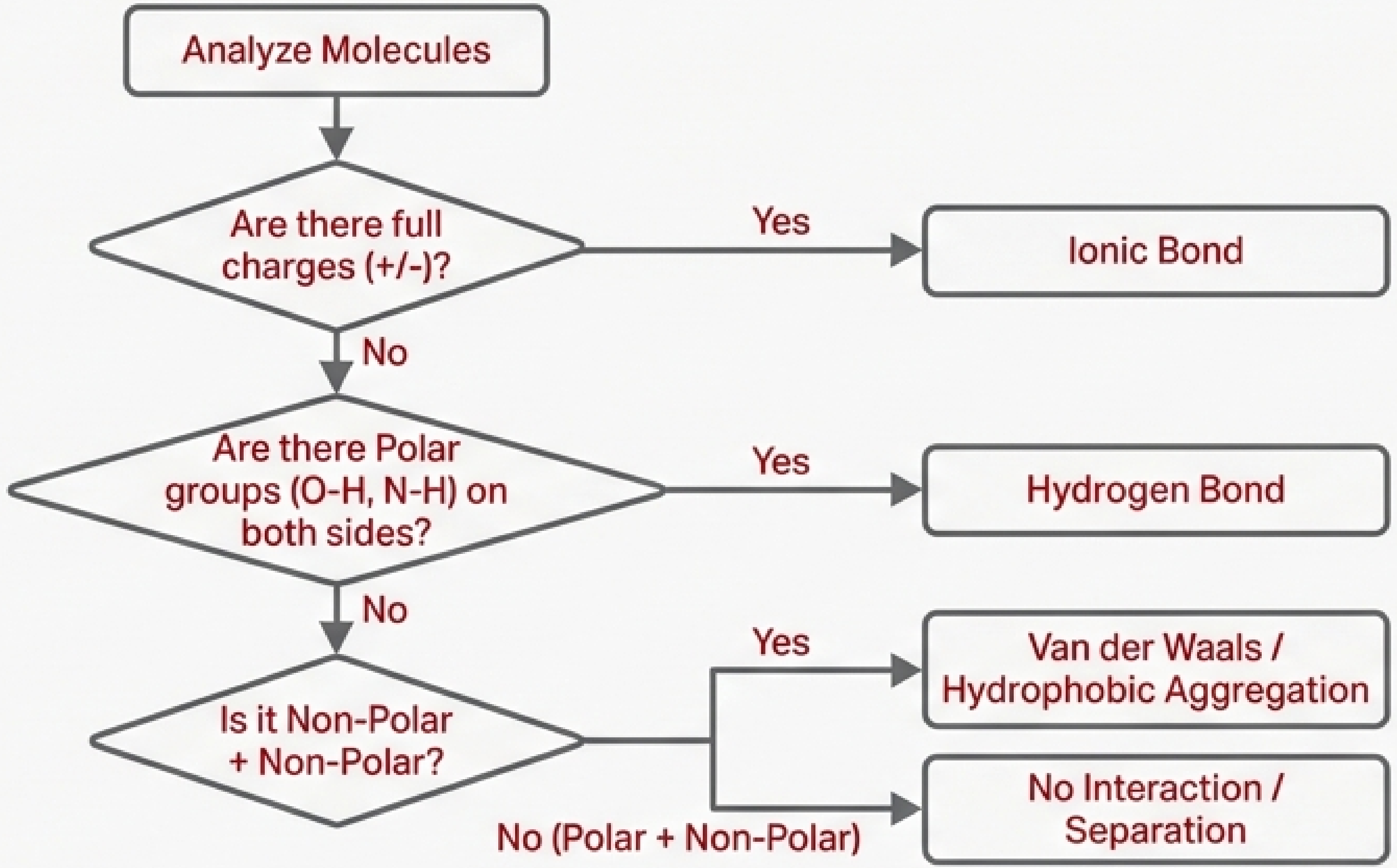


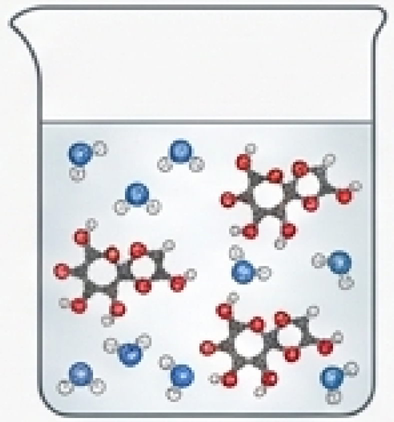
Fluctuating Dipoles
(~1 kcal/mol)

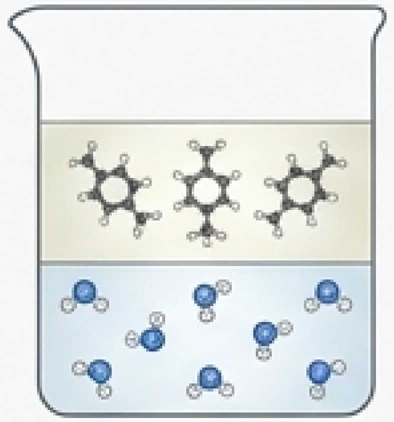
Strength in numbers: Geckos climb using millions of Van der Waals interactions.

(Crimson Pro)

Problem Solving: Predicting Interactions

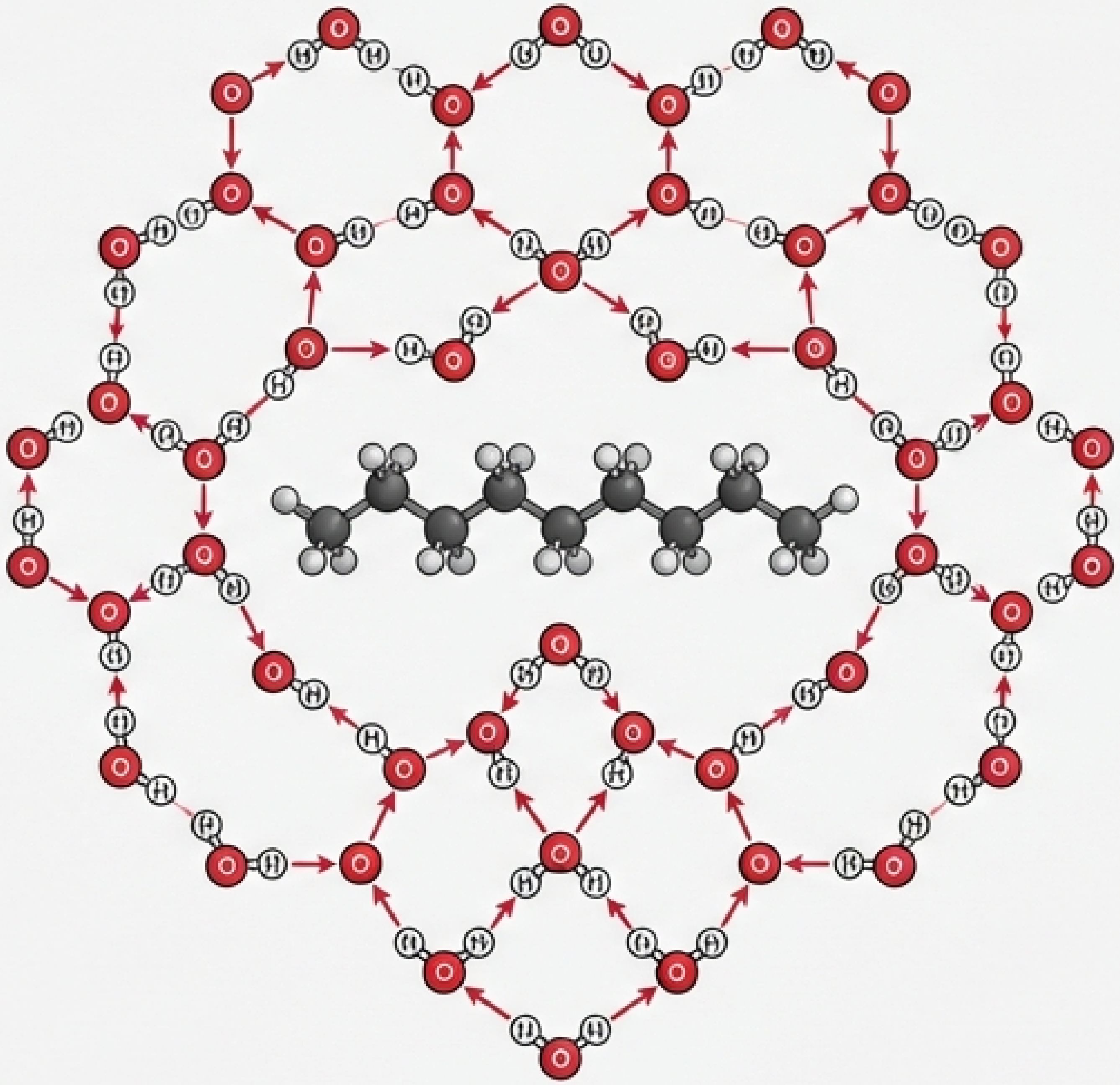



Sucrose + Water
(Dissolved/Homogenous)


Limonene + Water
(Separated layers)

Water & The Hydrophobic Effect

The Cage



Mechanism of Exclusion

1. Water maximizes its own hydrogen bonds (Entropy).
2. Non-polar molecules disrupt this network.
3. Water pushes non-polar molecules together to minimize their surface area.

The Hydrophobic Effect is driven by water's preference for itself, not an attraction between oil molecules.

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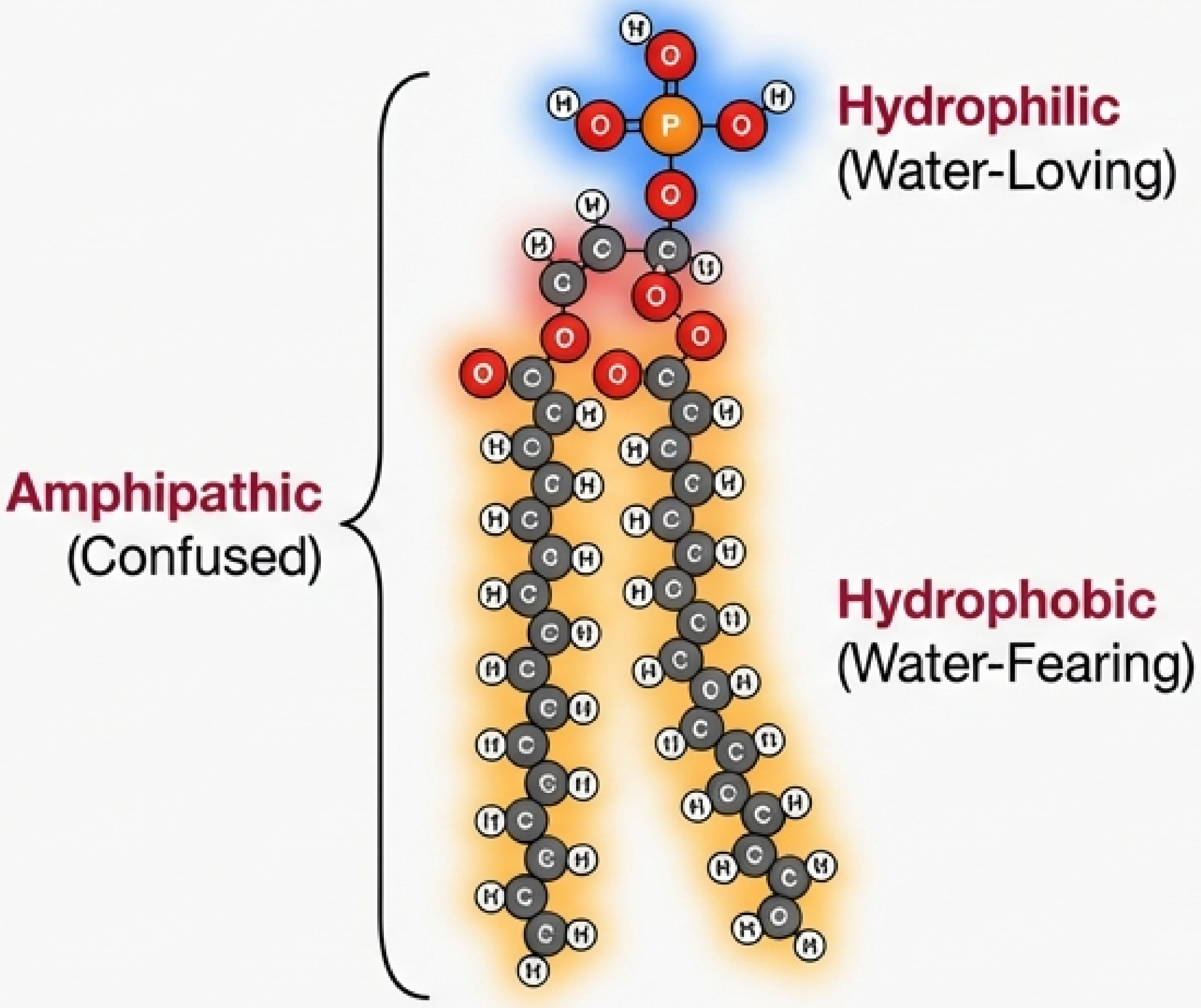
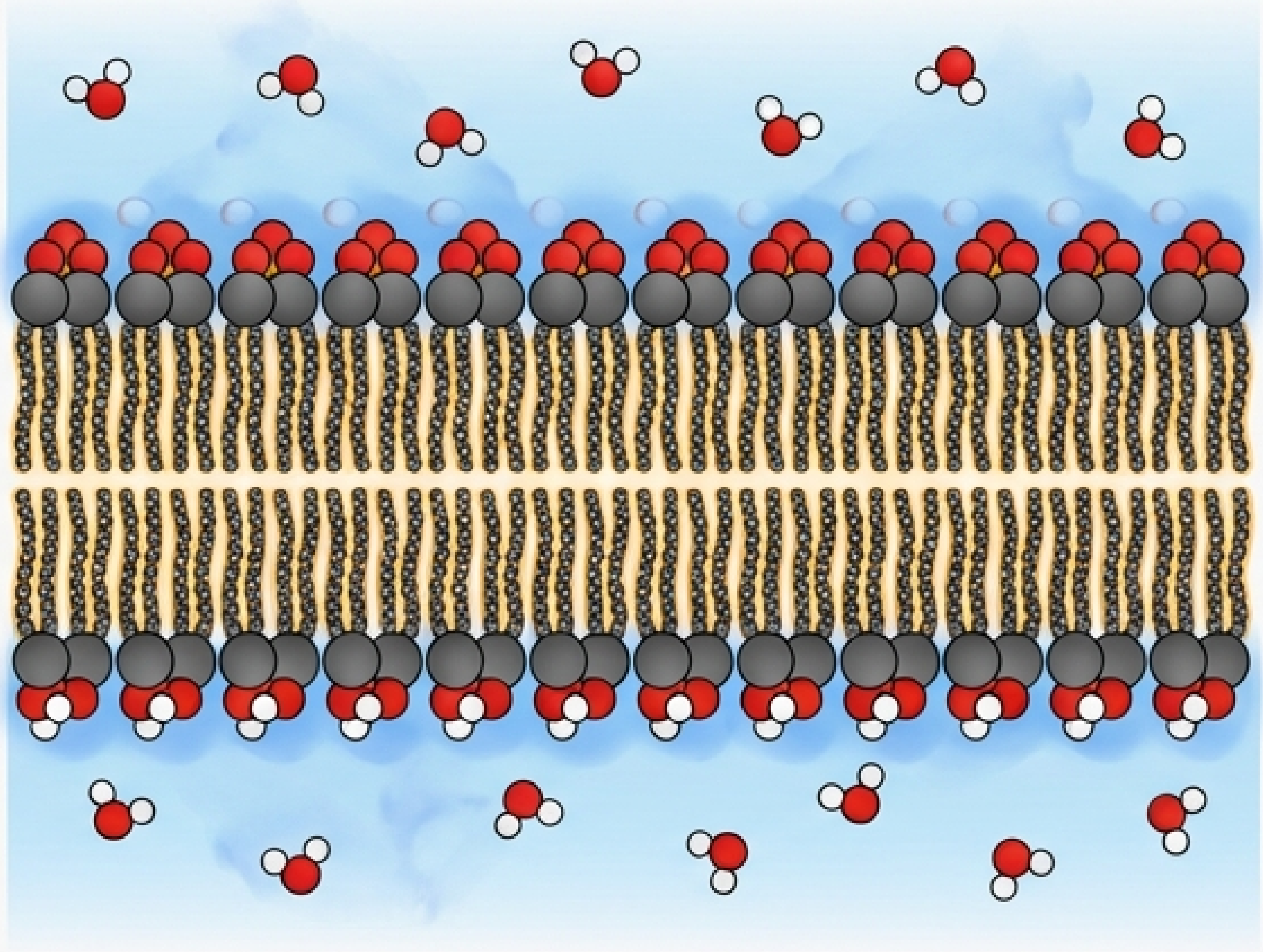
Emergence: Self-Assembling Membranes

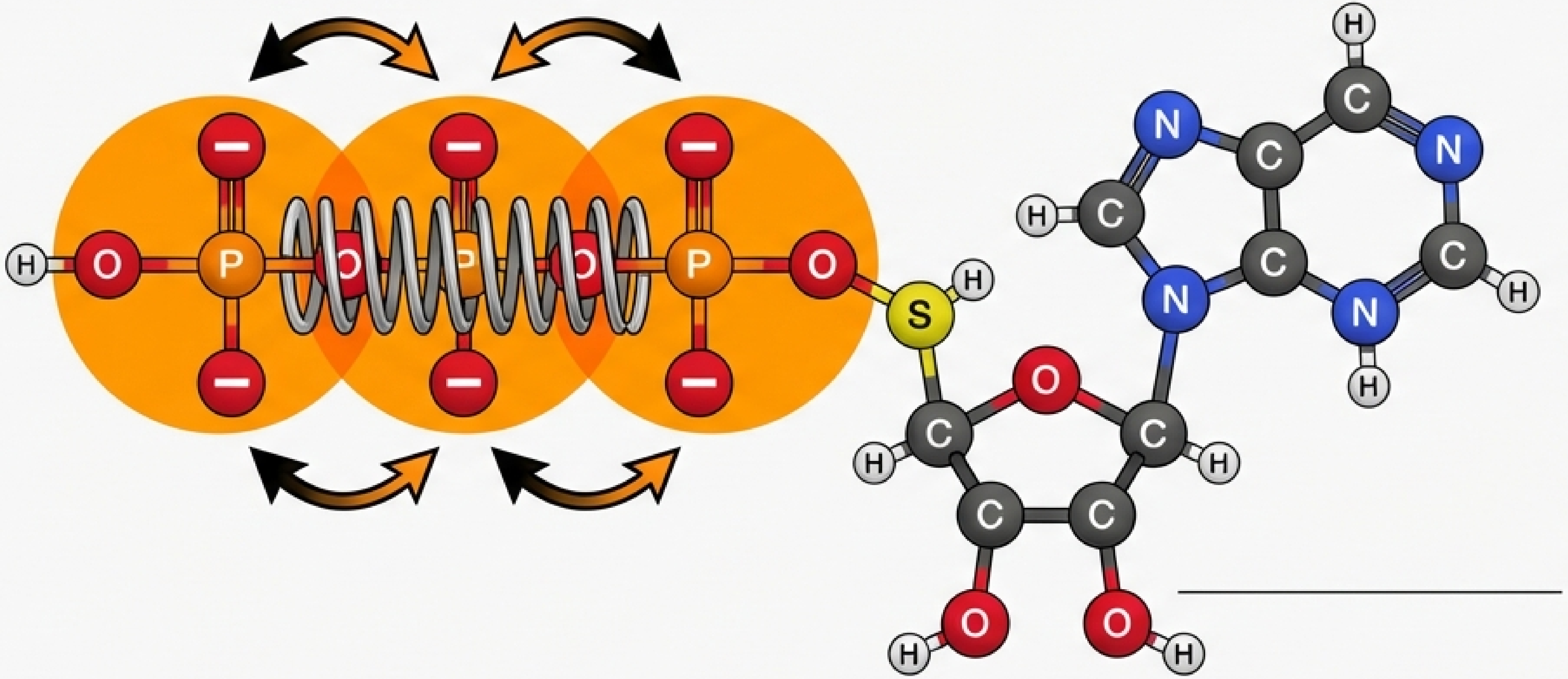


Phospholipid



Lipid Bilayer: Spontaneous assembly to hide hydrophobic tails.

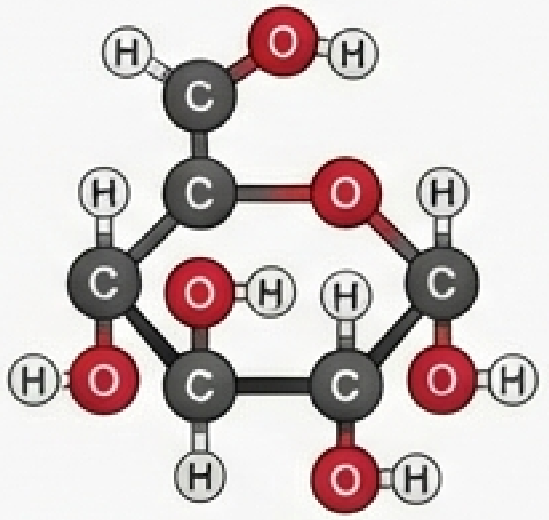
Emergence: The Energy Spring (ATP)



Potential Energy: Like a compressed spring, the 3 negative phosphates repel each other. Covalently bonding them together stores massive energy.

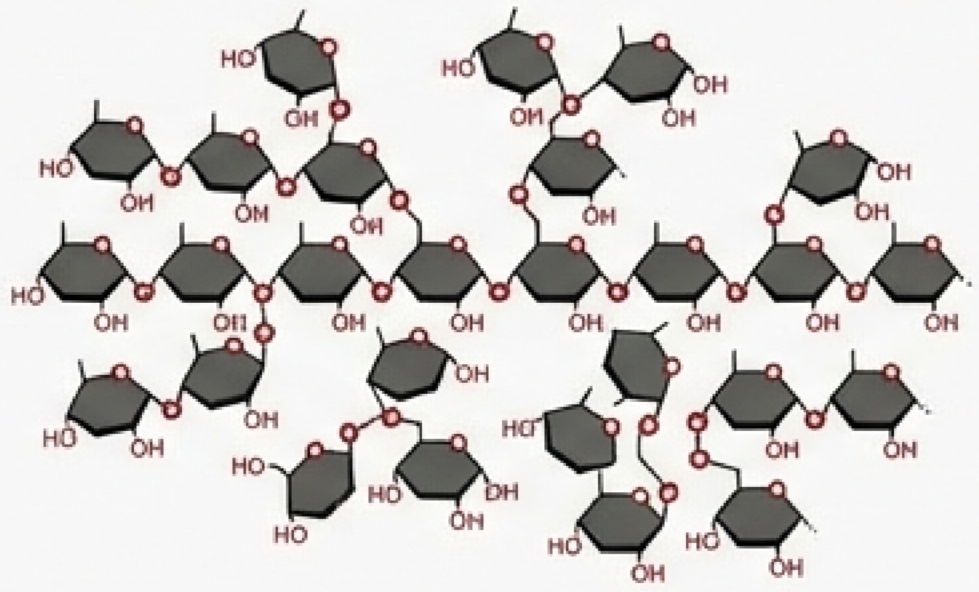
Hydrolysis = Releasing the spring.

Emergence: Carbohydrates & Structure



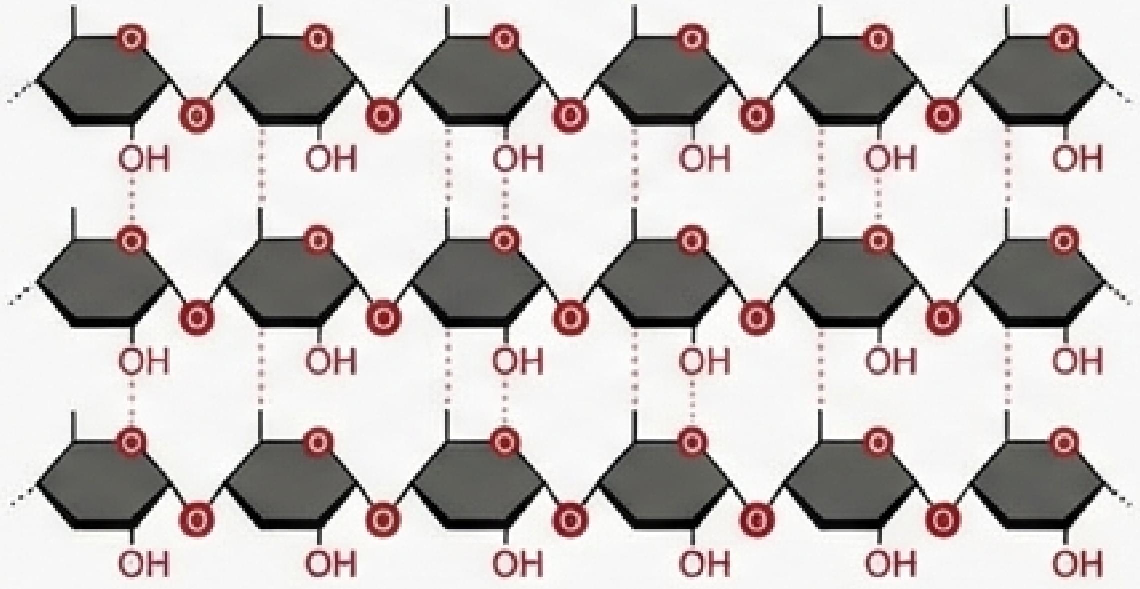
Monomer: Glucose ($C_6H_{12}O_6$)

Glycogen (Energy Storage)



Branched for rapid access.

Cellulose (Structure)



Linear and stacked for rigidity (Wood).

Connectivity determines function.

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The Biochemist’s Reference Matrix

Bond Type	Relative Strength	Basis of Interaction
Covalent	Strong (~80 kcal)	Shared Electron Pairs
Ionic	Moderate	Full Charge Attraction (+/-)
Hydrogen	Weak (~5 kcal)	Polar 'Sandwich' (Partial Charges)
Van der Waals	Very Weak (~1 kcal)	Fluctuating Dipoles

Polar: Look for O-H, N-H, or Charge.

Non-Polar: Look for C-H or C-C bonds.

Amphipathic: Molecules with both (e.g., Phospholipids).

“Biochemistry eliminates the need for ‘Vitalism.’ Life is understandable as a series of chemical interactions.”